Early Detection of Alzheimer's Disease from Speech with Deep Learning

Master's Thesis Progress Report

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Abstract

Early detection of Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) is critical for timely intervention and effective disease management. Speech analysis has emerged as a promising, non-invasive, and cost-effective approach for detecting the cognitive decline associated with AD and MCI. This master's thesis aims to contribute to the field by focusing on modeling the conversational nature of the available datasets, a crucial yet often overlooked characteristic. We model the utterances of the conversation as nodes of a graph, capturing the complex interactions between the interviewer and the participant. Moreover, we propose a novel technique of introducing logarithmic-step skip edges to capture long-range dependencies with a minimal computational cost. We additionally introduce a patient-wide pooling method that effectively isolates and evaluates the patient's speech. In this report, we document our progress through the stages of data analysis and preprocessing, baseline experiments and experimentation regarding conversational modeling, deriving inspiration from the field of paralinguistics.

1 Introduction

Dementia is a progressive neurodegenerative disorder that causes irreversible brain damage, significantly impairing activities of daily living. It is estimated to affect over 57.4 million people, a figure that is expected to rise dramatically due to the aging population [1]. Alzheimer's Disease (AD) is the most prevalent form of dementia, accounting for 60-70% of all cases [2].

While there is no known cure for AD, early detection can greatly help patients manage their symptoms and improve the quality of life for them and their families. In particular, researchers highlight the need for detecting a cognitive decline stage known as Mild Cognitive Impairment (MCI) which may or may not progress to dementia. Speech analysis has emerged as a promising approach for detecting MCI, offering a non-invasive, cost-effective, and accessible means of assessment [3], [4].

The field has witnessed significant academic interest in recent years, characterized by frequent competitions and a growing body of literature. However, it continues to face substantial challenges that hinder clinical translation. Although many models report high evaluation metrics, performance levels remain insufficient for reliable clinical application. Moreover, these models are frequently trained and evaluated on small or outdated datasets, raising concerns regarding their generalizability. In this research proposal, we perform a comprehensive literature review of existing methodologies and the evaluation metrics achieved. We then identify gaps in the literature and missed opportunities, and finally propose a set of promising methodologies to address these limitations.

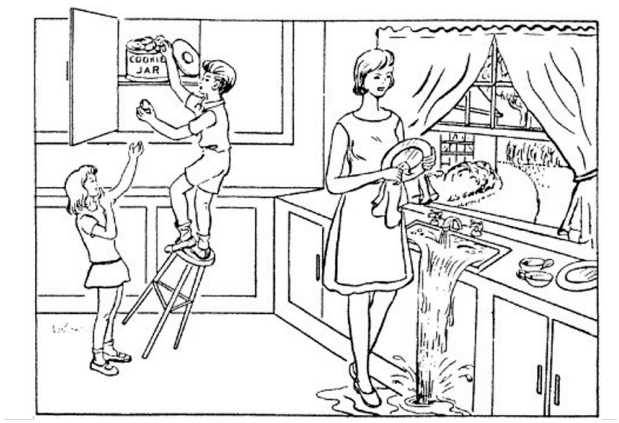

Figure 1: The Cookie Theft Picture, used in the picture description task of the Pitt corpus and other datasets from DementiaBank.

2 Relative Work

This study focuses on modeling the conversational structure of DementiaBank data, to assist the model in differentiating the interviewer from the participant, control each person's involvement in the model's decision-making and evaluate the participant's question answering skills in a more precise way.

2.1 Emotion Recognition in Conversations

Deriving inspiration from the field of Multimodal Emotion Recognition in Conversations (ERC) [5]–[7], we distinguish two possible directions for our methodology:

- **Prompt-based**: Having a foundational language model as a decision-making backbone, we engineer prompts to incorporate speaker text, speaker information, and/or speech embeddings.

- **Graph-based**: We model the conversation graph, with nodes representing speaker utterances and edges representing semantic connections. After being processed

After reviewing these approaches, we observed that *Graph-based* methodologies were proven the most promising for experimentation and improvement. In addition, at least in the domain of Emotion Recognition in Conversations, most state-of-the-art approaches incorporated the modeling of graph structures. Consequently, this is the path that we decided to focus our work on.

Originating in the RNN era of ERC, the first work to introduce the concept of modeling a conversation as a graph was DialogueGCN [8]. The authors modeled each utterance as a node, fully connected with its neighbors within a specific time window. Speaker information was encoded within the edges, defining a distinct relation for each interaction ($speaker_1 \rightarrow speaker_2$, $speaker_1 \rightarrow speaker_2$, etc.). MMGCN [9] expanded this idea into the multimodal setting by creating a separate graph for each modality and connecting inter-modal vertices of the same utterance. In contrast to DialogueGCN, they distinguished participants by concatenating a one-hot encoding vector into each vertex embedding. A recent improvement to this methodology was introduced by MERC-GCN [10]. The authors constructed a graph similar to that of DialogueGCN, but chose to fuse multimodal features with a co-attention transformer before representing each utterance as a single vertex. MERC-GCN's architecture is displayed in Figure 3. All of the aforementioned methodologies utilized LSTMs or GRUs for full context modeling prior to generating the graph, which they subsequently processed with GCNs.

2.2 Conversational Modeling in Other Domains

The modeling of conversations using graphs has been applied across various other domains in the broader literature. In the field of depression detection, [11] modeled clinical interviews from the DAIC-WOZ dataset using a dual-graph approach: one graph encoding sentence similarity and the other capturing keyword correlation. They also employed a multi-view methodology, creating distinct graphs for the interviewer's questions and the patient's answers, with explicit connections linking each question to its corresponding response. In a medical context, [12] developed a knowledge-grounded chatbot capable of offering advice to patients. For their conversational modeling, the authors represented utterances as nodes, linking adjacent utterances via neighbor edges. Furthermore, they extracted medical entities, referenced their relationships within an external knowledge graph, and introduced corresponding knowledge-aware edges. This approach allowed the authors to effectively target medical information and capture long-range dependencies. Finally, [13] utilized graphs to model dialogues centered around an initial question, aiming to extract the final answer. Instead of simply connecting adjacent utterances, they extracted Conversational Discourse Structures to define the relationships among them. Additionally, they represented participants as distinct nodes connected to their spoken utterances. Lastly, they introduced a global question node and connected it to all utterance nodes within the graph.

3 Dataset

Most available data in the field is provided by the *DementiaBank* [4] database. DementiaBank consists of a collection of corpora, that contain audio recordings and CHAT transcripts of various cognitive assessment tasks, administered by an interviewer[1]. In this work, we utilize the *MultiConAD* [14] framework, which aggregates all DementiaBank corpora into a single, unified dataset. In addition, MultiConAD converts all transcripts to plain text and experiments with translation and multilingual trainings. We choose to focus only on the purely English subset of MultiConAD.

Implementation Details: While the MultiConAD reproduction source code is available online[2], it was realised that the repository was missing critical preprocessing steps that were essential for the incorporation of the "WLS", "VAS" and "Kempler" corpora. Thus, these scripts had to be manually implemented. In addition, since the publication of the MultiConAD dataset, the "Delaware" corpus has grown in number of data samples, that we chose to incorporate into our implementation. As

[1]The only exception is the VAS corpus, which consists of recorded Voice Assistant System commands.
[2]https://github.com/ArezoShakeri/MultiConAD

a result, our MultiConAD implementation differs slightly from the original one. The differences between the two implementations are summarized in Table 1.

MultiConAD	Tot Length	Samples	Train	Test
Original	*not specified*	2411	2201	210
Ours	114.49 hours	2469	2217	252

Table 1: Differences in data samples between ours and the original (English only) MultiConAD dataset implementation.

After the implementation and analysis of the dataset at hand, several key observations have emerged that will shape the direction of this study's methodology:

1. **Conversational Data**: As already mentioned, most data samples consist of oral conversations. To our knowledge, no recent study has modeled this conversational structure, leading to models processing the speech of both the interviewer and the participant with no distinction whatsoever (see 6.3.3 in the Literature Review). This is where choose to focus our study.

2. **Small Dataset**: Although it includes nearly all available datasets, MultiConAD is still severely small for deep learning standards, amounting to about 2500 samples, or about 115 hours. Thus, the resulting model will be limited to fine-tuning, or to utilizing layers of open, pre-trained language and paralinguistic models.

3. **Class Imbalance**: As shown in Figure 2 (top and bottom right), the class distribution heavily favors the healthy controls. This raises the need for class-balancing data augmentation.

4. **Large-Scale Recordings**: The DementiaBank dataset consists of substantial audio files with significant variance in duration, ranging from 2 to 35 minutes. Given that most speech models are optimized for short windows of a few seconds, we have introduced a design requirement to manage the aggregation of these processed segments into a cohesive output. This can be better demonstrated by Figure 2 (top left).

5. **Dataset Bias**: The individual sub-datasets within MultiConAD exhibit a disproportionate class distribution (see Figure 2, bottom left). For instance, the WLS dataset comprises a vast majority of healthy controls. This imbalance introduces a significant bias hazard; the model may inadvertently learn to classify samples based on the unique acoustic signatures or noise fragments of the source datasets rather than the underlying pathological voice characteristics.

4 Methodology

New

In this section, we describe our proposed model and its computational stages, as well as our training strategies. We model the utterances of the conversation as nodes of a graph, capturing the complex interactions between the interviewer and the participant. Moreover, we propose a novel technique of introducing logarithmic-step skip edges to capture long-range dependencies with a minimal computational cost. We additionally introduce a patient-wide pooling method that effectively isolates and evaluates the patient's speech.

4.1 Architecture

Preprocessing: Our data pre-processing pipeline will be heavily inspired by Emilia-Pipe [15], and can be summarized as follows: Source Separation → Speaker Diarization → Audio Chunk Segmentation → Audio-Transcript Alignment. We distinguish source separation as a critical part of our pipeline, as each sub-dataset contains its unique noise fragments, which intoduces significant bias risks. regarding speaker diarization and chunk segmentation, we can utilize the available hand-crafted CHAT transcriptions, which involve speaker identification and accurate timestamps for utterances. In sub-datasets that CHAT transcriptions are not available, we will follow Emilia-Pipe's toolkit. Lastly, we choose to omit Voice Activity Detection, as periods of pause and silence are a strong indicator of cognitive impairment.

Feature Extraction: Each utterance contains two modalities: text and audio. For audio modality we will extract paralinguistic embeddigs utilizing Google's TRILLsson-5 model [16]. For the textual modality, a sentence transformer will be utilized.

Any suggestions on a modern SBERT-like model?

The next stages are reminiscent of the MERC-GCN architecture and the broader methodology used in the MERC domain. Thus, we display MERC-GCN's architecture in Figure 3, which we will be useing as a theoretical baseline for comparison.

Multimodal Fusion: For multimodal fusion, we will follow MERC-GCN's paradigm and fuse the audio and text embeddings before the graph modeling with a Co-Attention Transformer.

Graph Modeling: Similar to MERC-GCN, we propose modeling each utterance as a graph vertex, fully

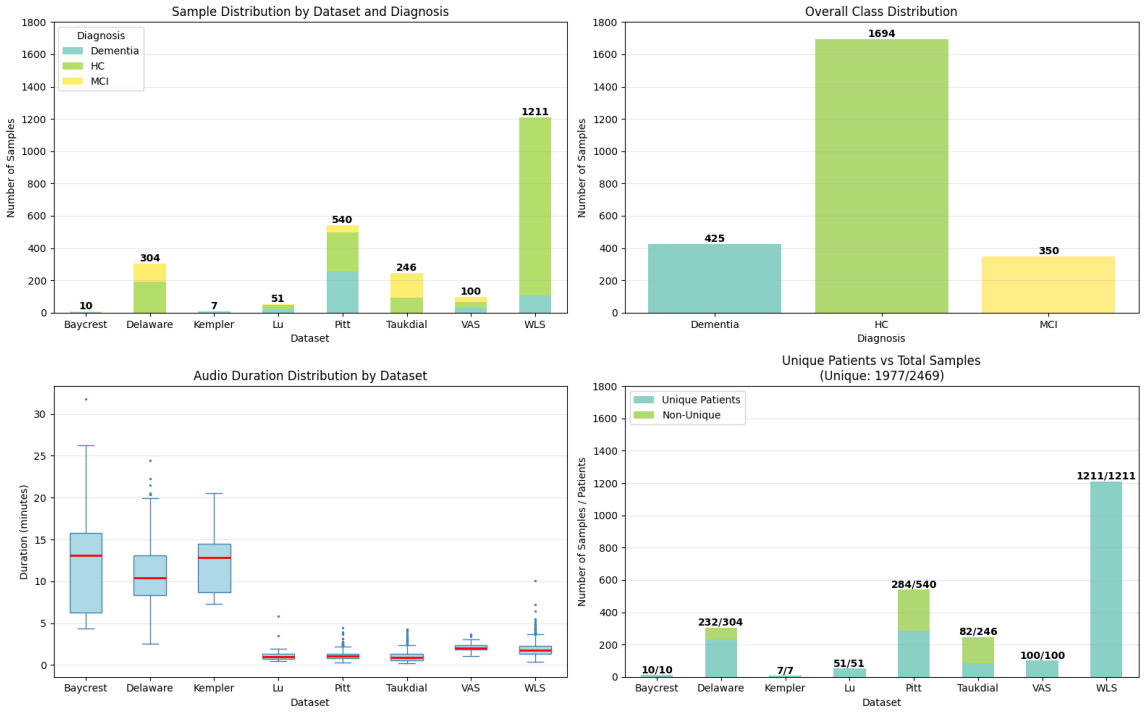

Figure 2: *top left:* Sample and class distribution among sub-datasets of MultiConAD. *bottom left:* Recording durations among sub-datasets of MultiConAD. *top right:* Class distribution. *bottom right:* Number of unique patients per dataset.

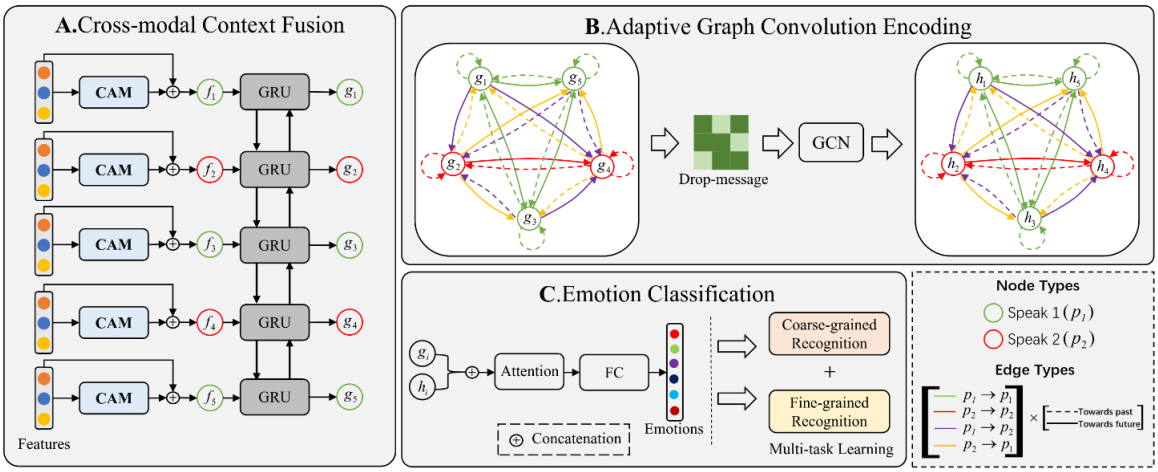

Figure 3: MERC-GCN architecture [10]

connected to its neighbors within a specific time window. However, in contrast to emotion recognition, capturing specific long-range dependencies plays a crucial role in dementia detection. Simply maximizing the context window within the graph is not a viable solution, as it renders computations prohibitively expensive [8]. Therefore, we propose a novel approach for capturing these long-range dependencies:

1. We omit the initial GRU-based contextual modeling utilized in MERC-GCN.

2. We retain the standard graph structure featuring fully connected local context windows.

3. For each utterance i, we introduce edges pointing to past and future utterances at logarithmic intervals (e.g., establishing connections to i±2, i±4, i±8, i±16, ...).

4. To accurately encode utterance ordering, we incorporate positional embeddings into the vertex representations.

We hypothesize that under this modeling scheme, two to three GNN layers will be sufficient to aggregate information across the entire sequence. Regarding the actual GNN model, we are considering utilizing a RelationalGAT instead of MERC-GCN's attention+RGCN approach. Since we have only 2 speakers, we are limiting the ralations to just 8, which computationally allows the use of different Q-K-V transformations for each relation i.e. an RGAT.

The authors of MERC-GCN include future and past relations. However, since we use positional embeddings, I assume it is best to remove this type of relation, reducing their total number to 4.

Pooling & Classification: For the classification step, we propose performing attention pooling over all patient utterances to extract a patient-wide representation, which is then applied to the final classification. This selective process prevents the model from directly classifying interviewer speech and instead focuses on the participant's speech and its interactions in the dialogue.

New

4.2 Parameter Estimation

In order to examine the sufficiency of the dataset, we perform a rough estimation of the proposed model's total parameter count. For each trainable stage of our architecture, we derive the theoretical formulas that give us the expected parameter count (biases are excluded for simplicity):

$$\begin{aligned} \text{projection} &: d_t d_h + d_a d_h \\ \text{co-attention} &: (3\frac{d_h^2}{h} \times h + d_h^2 + 2d_h) \times 2 \times L_{ca} \\ \text{RGAT} &: (d_h^2 + 3d_h^2) \times R \times L_{rgat} \\ \text{pool+classification} &: 2d_h^2 + 3d_h \end{aligned}$$

Where:

$$\begin{aligned} d_t &: \text{text dimention (sentence transformer)} \\ d_a = 1024 &: \text{audio dimention (TRILLsson-5)} \\ d_h &: \text{network hidden dimention} \\ h &: \text{co-attention heads} \\ L_{ca} &: \text{co-attention layers} \\ L_{rgat} &: \text{RGAT layers} \\ R = 8 &: \text{number of relations} \\ C = 3 &: \text{number of classes} \end{aligned}$$

Substituting the parameter values, we get a range of model size, from approximately **1.5m parameters** (for $d_t = 1024, d_a = 1024, d_h = 128, R = 8, C = 3, n_c a = 1, n_r gat = 2$), up to about **30m parameters** (for $d_t = 1024, d_a = 1024, d_h = 512, R = 8, C = 3, n_c a = 2, n_r gat = 3$). For further reduction of the trainable parameters, we would need to shrink the hidden dimension below 128, which we fear will lead to weak model generalization.

4.3 Training

Our proposed architecture introduces a sizable amount of trainable parameters. Direct model training on MultiConAD poses a significant risk of overfitting. Consequently, we'll have to first pretrain our model in a suitable dataset. Dataset candidates are shown below:

1. **Emilia**: A large dataset, scraped from internet sources such as podcasts and videos. Its English subset contains over *46,828 hours*. It should be noted, however, that it is mostly focused on TTS application and graph modeling will require some manual preprocessing.

2. **DeepDialogue**: Offers *40,150* automatically generated dialogues, with about *6 turns* per conversation, amounting to more that *480 hours* of audio.

3. **CANDOR Corpus**: A multimodal corpus of *1656* English dialogues that amounts to *850 hours* of audio. Each conversation lasts more than *25 minutes*, matching the long recording format of MultiConAD. Access to the dataset is granted after a formal request.

4. **Seamless Interaction**: Meta's newly open sourced conversational dataset, featuring more than *4,000 hours* of dialogues between *4,000+ participants*.

What pretraing algorithm should we use?

In the pretraining phase, the entire model will be trained end-to-end, which may include fine-tuning the unimodal encoders (the 88.6M-parameter TRILLsson and the yet-to-be-determined sentence transformer). Subsequently, during the MultiConAD training stage, only the GNN layers will remain trainable in order to mitigate the risk of overfitting.

Pretrain: finetune encoders in adition to model?
Finetuning: train only GNN instead of entire model?

5 Experiments

In this section, we outline our baseline progress, candidate experiments, and model results. We utilize Unweighted Average Recall (UAR) as our primary evaluation metric because it accounts for class imbalance and emphasizes false-negative classifications, which are critical in medical applications. Additionally, we report accuracy to facilitate direct comparison with the original MultiConAD baseline, alongside the Macro-F1 score due to its widespread adoption in the literature.

5.1 Baseline Experiments

The original MultiConAD paper proposes a series of simple baseline models that utilize the transcripts' text representations, in conjunction with traditional ML classifiers. For text representations, the authors experimented with TF-IDF representations (sparse) and `intfloat/multilingual-e5-large` embeddings (dense). Since our implementation of MultiConAD differs slightly from the original, we chose to reproduce these benchmarks to ensure a fair comparison. In Table 2, we present the best performing baseline classifiers, for sparse and dense representations.

Model	Acc	UAR	F1
Sparse+RF (original)	0.65	-	-
Dense+SVM (original)	0.65	0.63	-
Sparse+SVM (ours)	0.7	0.66	0.68
Dense+SVM (ours)	0.67	0.63	0.65

Table 2: Baseline model evaluation metrics.

[3] https://researchers.wls.wisc.edu/

New

Our evaluation demonstrates a performance increase of up to +5% compared to the original MultiConAD baseline. To investigate this discrepancy, we systematically ruled out several variables. We utilized the authors' original source scripts and default random seeds, eliminating architectural differences. To account for dataset size and split composition, we reduced our sample size to 2411 and ran train-evaluation loops across 10 iterations using random train-test splits. Under these conditions, the model consistently outperformed the original baseline, achieving acc: 0.690 ± 0.027, UAR:0.664 ± 0.026, and Macro−F1:0.675 ± 0.028.
Further analysis revealed noticeable variations in the sample distributions across sub-datasets (we estimate about 345 different samples). For instance, our implementation yielded fewer WLS samples but more Delaware and Taukdial samples than the authors reported. Because the original paper describes the pipeline but omits specific sample IDs for the train and test splits, achieving an exact 1-to-1 reproduction of their dataset seems unfeasible. We therefore conclude that the performance differences do not result from specific train-test split seeds, but rather stem from fundamental differences in sample distribution between the two implementations.

Exact reproduction of the MultiConAD dataset, particularly its specific test set, appears unfeasible. Furthermore, the original authors' train-test split methodology fails to account for speaker IDs, which introduces the risk of data leakage. Given these limitations, how should we proceed? Should we contact the authors?

6 Notes

- Although MultiConAD provides 1211 artificially labeled WLS recordings, the total amount of WLS samples provided by DementiaBank amount to approximately 10000 recordings from about 1300 speakers. Moreover, even more data samples may be retrieved by contacting the official WLS study organization[3]. This opens the door to potential self-supervised or silver-labeling techniques as a means to mitigate the limited dataset size issue.

Appendix: Literature Review

6.1 Datasets and Evaluation Challenges

DementiaBank: Most datasets currently available are provided by the DementiaBank database. DementiaBank is a large collection of audio recordings, consisting of various cognitive assessment tasks, administered by an interviewer. Most of these recordings are manually transcribed in the CHAT format, a transcription method that incorporates various linguistic observations such as pauses and stutters, speaker roles, and timestamps along with the spoken words. The most widely used of the provided datasets is the Pitt corpus, which includes mostly the "Cookie Theft Picture Description" task, where participants are asked to describe the image shown in Figure 1. There has also been work toward the creation and expansion of the Delaware corpus, a new dataset that focuses on the binary MCI vs HC distinction. It provides a large variety of cognitive tasks and includes subjects of a wide ethnic and cultural diversity [4].

WLS: The WLS is a large-scale, extended longitudinal study of a random sample of 10,317 men and women who graduated from Wisconsin high schools in 1957. The WLS participants were interviewed up to 6 times between 1957 and 2011. DementiaBank provides access to a subset of audio recordings from the 2003 and 2011 interview rounds, amounting to about 1300 different speakers. Although this is a dataset of substantial size, it lacks clinical labels for cognitive impairment, limiting its direct applicability to supervised learning tasks.

ADReSS, ADReSSo and ADReSS-M: The ADReSS Challenge was introduced in the Interspeech 2020 conference. It provides a balanced subset of the Pitt corpus, with respect to age and gender [17]. The ADReSSo Challenge followed in 2021, introducing a more difficult task of AD detection using only speech samples, without manually created transcriptions. It also utilized a subset of the Pitt corpus [18]. Lastly, the ADReSS-M challenge, introduced in the ICASSP 2023 conference, focused on the binary classification task of AD detection in the Greek language. It provided a subset of the Pitt corpus, as well as a new dataset with Greek speech samples [19]. Even after the end of these challenges, the datasets remain available and have been widely used as benchmarks for new approaches.

MultiConAD: The MultiConAD dataset [14] is a recent effort to tackle the problems of 3-class classification and multilingual generalization. It combines multiple existing datasets, mostly the ones provided by DementiaBank, to create a large and diverse multilingual dataset, with a variety of cognitive assessment tasks. It includes audio and transcription samples in English, Spanish, Chinese, and Greek. Additionally, it provides an important set of baseline models that act as a starting point for future research.

PROCESS: The ICASSP 2025 PROCESS Grand Challenge [20] introduced a modern dataset to serve as a benchmark for the 3-class detection problem. It provides audio recordings and manual transcripts (with only audio provided for the test set) from three cognitive tasks: Semantic Fluency, Phonemic Fluency, and Picture Description. The contestants were also tasked with performing regression on the MMSE score, a widely used clinical assessment for cognitive impairment.

6.2 AD and MCI Detection from Speech

In this section, we summarize the general trends in methodologies tackling AD and MCI detection from speech. We organize our review by general methodologies; however, these subsections loosely resemble the chronological development of approaches.

6.2.1 Feature Engineering

Initial work emphasized the extraction of information-rich features, used in conjunction with traditional ML methods. These approaches proved very effective in the challenge settings as well, where the applications were limited to the small challenge datasets. These methods extract mostly hand-crafted acoustic (e.g., pauses, low-level descriptors, etc.) and linguistic (e.g., verbal richness, filler words, etc.) features and/or semantic representations, mostly from the BERT [21] language model.

Such approaches have led to winning submissions in the ADReSS and ADReSSo Challenges [22], [23]. In the more recent PROCESS Challenge, the overall winner leveraged linguistic features, extracted from ASR transcriptions, and an ensemble of traditional machine learning models, to achieve an F1 score of 0.649. These linguistic features include cognitive-task-specific indicators (count of correct words in Verbal Fluency Tasks) and algorithmically extracted speech pause descriptors [24]. The best performing submission specific to the classification task achieved an F1 score of 0.696, using a self-developed Digital Linguistic Biomarker (DLB) extractor [25].

In a notable paradigm of this methodology, the authors of ADscreen [26] extract a broad range of features, including metrics related to phonetic motor planning, semantic disfluency (e.g., word repetition and pausing), lexical diversity, and syntactic structure. They further incorporate BERT embeddings to capture verbal disfluencies, as well as psycholinguistic features such as LIWC and GeMAPS. Similarly, Haider et al. focus on the emotional dimension of speech by introducing the Affective Behaviour Representation (ABR), which employs a machine learning model to label each speech segment with an emotion and summarize the emotional content of an entire recording into a single vector [27].

6.2.2 Deep Learning Architectures and Multimodal Fusion

While feature engineering relies on extracting predefined markers, deep learning approaches aim to learn latent representations directly from the data. Given the success of deep learning in the more general field of paralinguistics, researchers naturally sought to implement it in the AD detection from speech domain. Deep learning architectures were used to produce meaningful latent representations and fuse acoustic and textual cues.

Within early works, Liu et al. [28] feed MFCCs into a DNN encoder to produce low-dimension bottleneck features of 40ms time frames, followed by CNN and BiLSTM layers. Evaluating their model with 10-fold CV on the Pitt corpus, they achieved an accuracy of 82.59% and an F1 of 82.94%.

In the multimodal setting, a wide variety of methods and architectures emerged. These approaches experiment with various combinations of features, such as BERT embeddings, traditional acoustic features, and other neural-network extracted features [29]–[33]. In an interesting approach, Lee et al. achieved state-of-the-art results in the ADReSSo Dataset by processing the Cookie Theft Picture description task image with VLMs and comparing it with the patient's text modality to check the validity of the response. They additionally utilized the Shapley value from game theory to introduce a new auxiliary loss function, which includes information about each modality's contribution [34].

6.2.3 Fine-tuning and Foundational Models

Foundational models have been influencing the domain of AD detection from speech since the early days of the ADReSS Challenge. As mentioned in the preceding paragraphs, BERT was frequently used to extract semantic embeddings that were later used as features for classification. Today, Large Language Models are still widely used [35]–[37], utilizing ASR transcriptions from state-of-the-art models and leading to promising results. An extensive study that compared various text-based ML methods showed that the fine-tuning of pretrained models is the most performant approach [38], at least in the text-only setting.

However, limiting the input to text only discards very crucial biomarkers embedded in the patient's voice. While some have resorted to encoding audio information into the transcripts [35], others have fine-tuned the ASR models themselves to allow the processing of purely acoustic inputs [39]–[41].

Most recent works merge acoustic and linguistic information by incorporating the use of Large Audio-Language Models [42], [43]. Zolnour et al. use a late fusion architecture, combining the predictions of a classifier that is fed pretrained encoder features, and a classifier that is fed handcrafted linguistic features. They also perform extensive experimentations with unimodal and multimodal LLMs, fine-tuned on classification from text and audio. Lastly, they generate synthetic text by prompting foundational LLMs. Overall, the late fusion scheme outperformed the fine-tuned LLMs [42]. Shanin et al. bypass fine-tuning entirely by simply prompting an Audio-Language model. Although the results were not optimal, they were comparable with supervised methods [43].

Lastly, regarding intermediate pretraining, Zhu et al. [44] used language model perplexity metrics to select large datasets from the GLUE benchmark that are maximally similar to the ADReSS Challenge dataset. Then, they pre-trained a language model on the chosen datasets, followed by fine-tuning on the AD detection task. They additionally utilized their used perplexity-based metrics to invent a new sample-level pretraining technique, where samples that don't reduce perplexity are discarded.

6.2.4 Data Augmentation

Despite the known issue of data scarcity in the field, the applications of data augmentation are limited. To our knowledge, only two publications in the field study explicitly the creation of synthetic data. The authors of CDA [45] propose a contrastive data augmentation technique that simulates cognitive decline by removing random words from sentences, and generates positive samples with multiple passes in conjunction with dropout. Hlédiková et al. [46] perform extensive experimentation with various data-space augmentation techniques. The methods included classical acoustic and verbal perturbations and deep learning-based ones, namely voice conversion with the FragmentVC model, lexical paraphrasing with the Pegasus model and text generation using GPT-2. The authors conclude that their tested neural-based methods perform similarly to traditional ones, still achieving significantly high results.

Some other works have incorporated data augmentation into their methodologies. Specifically, Liu et al. [28] utilize SpecAugment [47], Runde et al. [37] apply Synthetic Minority Over-sampling to balance out the datasets, and Lin and Washington perform Synonym Replacement for their text-based model. Additionally, the authors of LLMCARE [42] experiment with various LLMs to generate synthetic transcripts.

6.2.5 State Of The Art

In Table 3, we summarize the datasets that serve as benchmarks in the field, detailing their sample sizes and number of classes. Alongside each dataset, we report the most successful model we could locate in the literature and its corresponding performance metrics. However, we emphasize that the literature domain is currently quite fragmented. The standardized contest datasets are

limited in size, and larger alternatives often lack determined train-test splits or standardized evaluation methods (e.g., simple inference vs. cross-validation). This lack of standardization makes objective comparisons and the identification of a true SOTA challenging. Despite these limitations, this overview highlights the primary data resources available and the current state of performance.

All mentioned dataset, besides the Pitt Corpus and MultiConAD were introduced as evaluation challenges. ADReSS and ADReSSo are subsets of the Pitt Corpus. MultiConAD is an aggregated dataset combining multiple DementiaBank corpora. Although it is multilingual, we focus only on the English subset. Since MultiConAD is a very recent dataset, the only model for comparison is the baseline set by the authors.

6.3 Limitations and Opportunities

6.3.1 Generalizability Concerns

While the scientific community has achieved remarkable evaluation metrics and made impressive strides in the field, these advancements are currently limited by the small, often outdated and noisy available datasets—often containing only a few hundred samples. Consequently, the applicability of these models in a clinical setting must be questioned.

In a study by Runde et al. [37], the authors managed to achieve 0.99 accuracy and F1 on the Pitt Corpus - a superset of the ADReSS and ADReSSo datasets - by utilizing Wav2Vec transcripts and ada-002 text embeddings. However, when tested on a 10-fold cross validation scheme, these scores dropped to 0.79. A more concerning example involves researchers achieving nearly 100% accuracy on the Pitt Corpus, using solely the silent segments of the audio recordings. This demonstrates the presence of a "Clever Hans" effect, where models achieve high accuracy not by actually learning the underlying mechanisms of the problem, but through spurious correlations in the training data [53]. These findings strongly emphasize the need for a larger, more diverse dataset, a gap promised to be filled by MultiConAD.

6.3.2 Interpretability

Clinical adoption necessitates model interpretability, as healthcare professionals must understand which speech characteristics drive diagnostic predictions in order to integrate findings with other clinical indicators and establish professional trust. However, most current approaches utilize complex deep learning architectures and pre-trained language models that function as "black boxes," thereby obscuring the decision-making process. While the domain may not yet have prioritized interpretability as an immediate prerequisite, future methodologies must be designed with the understanding that clinical viability ultimately depends on the transparency and interpretability of the model.

6.3.3 Untreated Conversational Nature of Data

With the exception of the 100 samples of the VAS (Voice Assistant System) corpus, all of the DementiaBank data consist of recorded dialogues, where an interviewer administers a cognitive test with the patient. To our knowledge, no recent study has accounted for this conversational structure, instead treating the recordings as homogeneous samples. Consequently, models analyze semantic and paralinguistic information from both the patient and the interviewer indiscriminately. Previous studies have highlighted this limitation, demonstrating that the interviewer's speech significantly impacts observed linguistic features, thereby potentially introducing bias and confounding model predictions [54].

6.3.4 Addressing Data Limitations

The most significant constraint within the studied domain is the limited size of available datasets, which restricts methodological innovation and hampers the generalization capabilities of proposed models. Furthermore, the acquisition and dissemination of such data are inherently slow processes due to severe data privacy concerns. Consequently, it is expected that data availability in this field will consistently lag behind industry standards established for more general tasks. While advanced techniques for mitigating data scarcity–such as data augmentation, synthetic data generation, and advanced fine-tuning–should be central to research in this area, they appear to be underrepresented in the current literature. Moreover, the WLS dataset remains significantly underutilized. Despite being unlabeled, this dataset offers substantial potential for pre-training and silver labeling strategies prior to fine-tuning on labeled data. To the best of our knowledge, the only studies that have leveraged WLS in this manner are [14] and [55], which manually determine potential labels based on task performance criteria.

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Dataset	Train/Test Split	Classes	Modalities	Performance	Source
Pitt Corpus	552 total (10-fold CV)	2 (AD, HC)	Audio, CHAT Transcripts	**Acc**: 0.95 **F1**: 0.95 **AUR**: 0.93	[48]
ADReSS	Train: 108 Test: 48	2 (AD, HC)	Audio, CHAT Transcripts	**Acc**: 0.94	[49]
ADReSSo	Train: 166 Test: 71	2 (AD, HC)	Audio	**Acc**: 0.96 **F1**: 0.96	[50]
ADReSS-M	Train: 237 Test: 46	2 (AD, HC)	Audio	**Acc**: 0.87 **RMSE**: 3.73	[51]
TAUKADIAL	Train: 387 Test: 120	2 (MCI, HC)	Audio	**UAR**: 0.86	[52]
PROCESS	Train: 157 Test: Hidden	3 (AD, MCI, HC)	Audio, Transcripts	**Macro-F1**: 0.696	[25]
MultiConAD (English subset)	Train: 2201 Test: 210	3 (AD, MCI, HC)	Audio, Transcripts	**Bin Acc**: 0.90 **Ter Acc**: 0.65	[14]

Table 3: State-of-the-Art (SOTA) Models for Alzheimer's Disease Detection from Speech.

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